LENSKART DAILY TRANSACTION ANALYTICS

Order & Store-Scale Demand Analytics — Descriptive Analytics Report

Report Focus Keywords: Top Category Performance, Revenue Concentration, Volume Trends, Regional Demand Distribution, Channel Tier Allocation, Customer Segment Share, Payment Rail Preferences, Order Success Rate, Inventory-Stockout Risk, Descriptive-to-Predictive Transition.

Dataset:	Lenskart Daily Transaction Data	**Time Period:**	30 days (2026-07-31 to 2026-08-29)
Records:	60 stores x 10 categories x 8 regions	**Gross Transaction Value:**	~INR 252 Cr
Analytical Context	E-commerce Eyewear & Vision Care Operations	**Analytical Stage:**	Descriptive Analytics Report

Key Performance Indicators (KPI Snapshot)

Headline operating metrics for the 30-day observation window, computed directly from the aggregated transaction dataset.

1.38M	**INR 252 Cr**	**INR 1,821**	**97.4%**
Total Orders (30 Days)	Gross Transaction Value	Avg. Order Value (AOV)	Avg. Order Success Rate
9 / 60	**25.9%**	**13.8%**	**43.1%**
Stores at Inventory Reserve Risk	Top Category Value Share (Eyeglasses)	Top Region Value Share (North Zone)	Top Segment Value Share (Regular)

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Note on Transaction Data

A real e-commerce platform's live order-management transaction feed isn't publicly accessible, so this descriptive report is computed from a synthetic dataset built to mirror real eyewear & vision-care e-commerce transaction patterns: 60 stores across 10 product categories, 8 regions, over 30 days (2026-07-31 to 2026-08-29), totaling ~1.38 million simulated orders worth ~INR 252 crore. It is designed to show exactly what a descriptive analytics pipeline processing millions of daily e-commerce transactions would produce — the same structure applies directly once real order-management data is substituted in.

Transaction Data Completeness & Quality Audit

The order logs contain zero missing cells and strict data-type integrity across all fields (date, store, region, category, order count, value, success rate, inventory reserve). Each aggregated row represents one store / day / category combination, providing an ideal baseline for evaluating operational performance and customer purchasing behavior.

Research Overview & Analytical Framework

Descriptive analytics summarizes historical order records to answer “what happened?”, “who transacted?”, and “how much value moved?” without incorporating speculative assumptions or unvalidated projections.

Analytical Dimension	Operational Dataset Implementation	Commercial Interpretation
Category Performance	Grouping value/count by 10 product categories	Identifies highest-revenue and highest-volume categories
Weekly Volume Velocity	Summing counts and value per calendar week	Measures order throughput momentum
Order Value Spread	5 binned average-value tiers (store-day level)	Quantifies basket size from everyday to luxury/bulk
Channel & Segment Mix	Channel-tier and customer-segment aggregations	Measures infrastructure load and relationship value
Geographic Channel Reach	Region and macro-cluster aggregations	Tracks geographic demand concentration

Objective & Core Business Questions

The objective of this descriptive analysis is to extract empirical operational insights and commercial transaction patterns across categories, channels, segments, and geographies.

- **Question 1 (Top Categories):** Which product categories generate the highest value versus the highest order count? (Section 1)
- **Question 2 (Volume Trends):** How do order count and value fluctuate week over week? (Section 2)
- **Question 3 (Average Order Value):** What is the distribution and spread of order values across stores? (Section 3)
- **Question 4 (Channel Allocation):** Which channel tier delivers the optimal balance of volume and value? (Section 4)
- **Question 5 (Customer Segment Share):** How do customer segments impact order yields? (Section 5)
- **Question 6 (Regional Demand):** Which geographic regions drive the largest order volumes? (Section 6)
- **Question 7 (Payment Preferences):** Which payment rails are favored across high-value orders? (Section 8)
- **Question 8 (Reliability):** How are orders performing on success rate, and where is reliability slipping? (Section 9)

- **Question 9 (Inventory Risk):** Which stores are at risk of stockouts? (Section 10)

How This Works at Real Scale (Analytics Stack)

In enterprise e-commerce environments, descriptive analytics pipelines process millions of order-management, payment-gateway, and warehouse-management records daily using accessible, highly scalable tool stacks.

Architecture Layer	E-commerce Operations Purpose	Standard Data Science & BI Tools
1. Order Capture	Log checkout, payment-gateway and warehouse events in real time	Kafka / Kinesis, REST APIs
2. Data Cleaning & Wrangling	Validate line items, verify count x value, reconcile SKUs	Python (Pandas), SQL
3. Fraud / Risk Scoring	Real-time decline & anomaly scoring on each order	Rules engine + ML model serving
4. Dimensional Storage	Store order fact tables and store/region dimension tables	PostgreSQL, Cloud Warehouses
5. Metric Aggregation	Compute AOV, success rate, regional share and correlations	Excel Pivot Tables, Python, SQL
6. BI Reporting & Dashboards	Generate executive visual charts, KPI scorecards, and trends	Power BI, Tableau, Matplotlib, Looker

1.0 Top Categories & Gross Value Performance

Evaluating performance across the 10 recorded product categories indicates that **Eyeglasses (Prescription)** leads in gross transaction value, generating INR 65 crore (25.9% value share). **Sunglasses** follows with INR 45 crore (17.8% share). Combined, these two categories account for 43.7% of total transaction value.

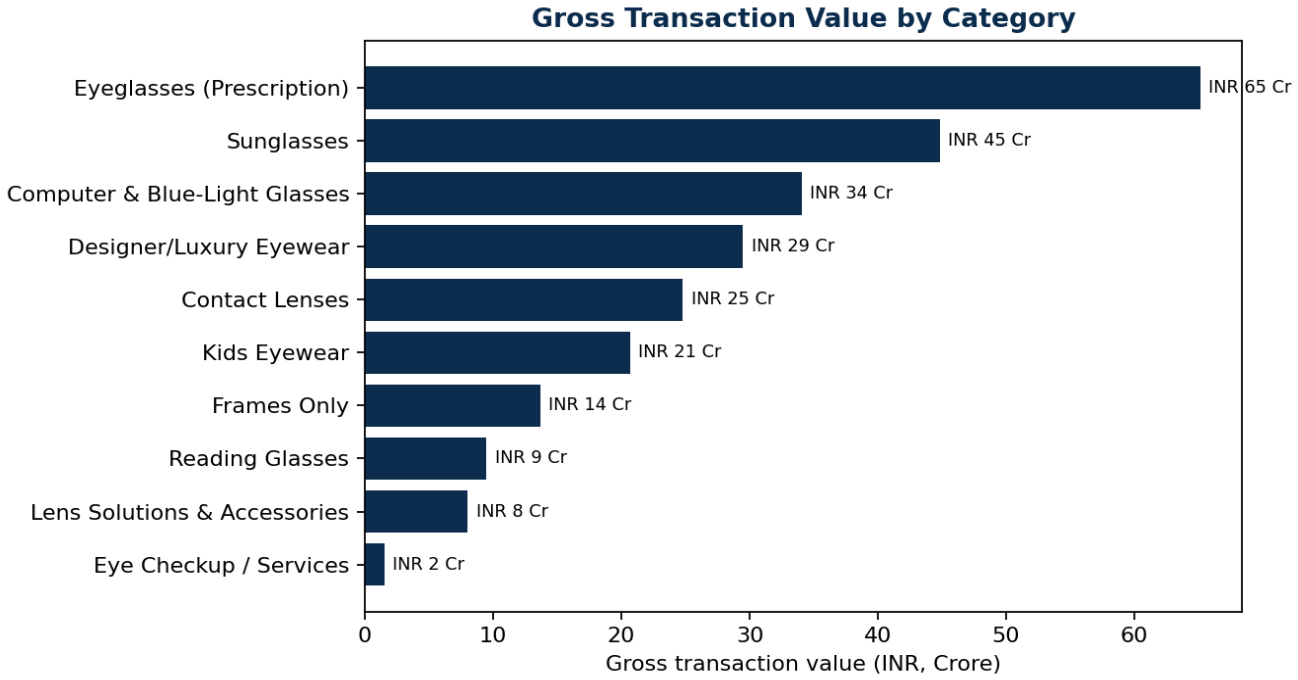


Figure 1. Gross Transaction Value by Category.

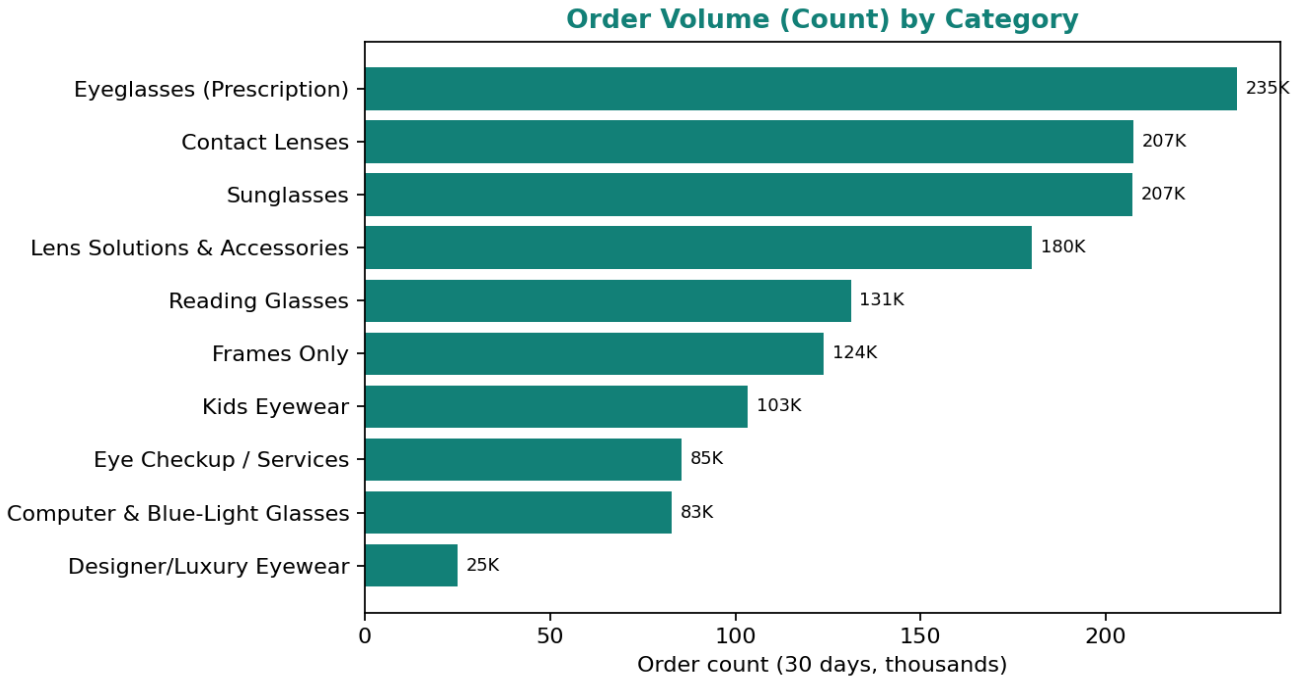


Figure 2. Order Volume (Count) by Category.

Category & Value Performance Table (30-Day Aggregate)

Category	Count	Value (INR Cr)	Value Share (%)	Avg Value / Order (INR)
Eyeglasses (Prescription)	235,401	65	25.9%	2,767
Sunglasses	207,151	45	17.8%	2,164
Computer & Blue-Light Glasses	82,855	34	13.5%	4,109
Designer/Luxury Eyewear	25,111	29	11.7%	11,744
Contact Lenses	207,336	25	9.9%	1,196
Kids Eyewear	103,385	21	8.2%	2,000
Frames Only	123,899	14	5.4%	1,103
Reading Glasses	131,144	9	3.8%	723
Lens Solutions & Accessories	179,993	8	3.2%	446
Eye Checkup / Services	85,473	2	0.6%	180

Deeper Insight — Value Leadership vs. Volume Leadership

The dataset exhibits a vital descriptive divergence between value leadership and count leadership. While Eyeglasses (Prescription) achieved the highest gross value, Eyeglasses (Prescription) led in physical order count (17.0% count share) — a small number of high-value, lower-frequency categories carry outsized weight in value terms even though they represent a smaller share of total count.

Commercial Recommendation
Operations and merchandising teams should track value and volume leaderboards separately: capacity planning (store staffing, lens-lab/glazing bandwidth) should follow the count leaderboard, while cash-flow and margin planning should follow the value leaderboard.

2.0 Weekly Volume Trends & Value Trajectory

Evaluating week-by-week order velocity highlights how payday cycles, festive/sale-event timing, and campaign scheduling affect throughput.

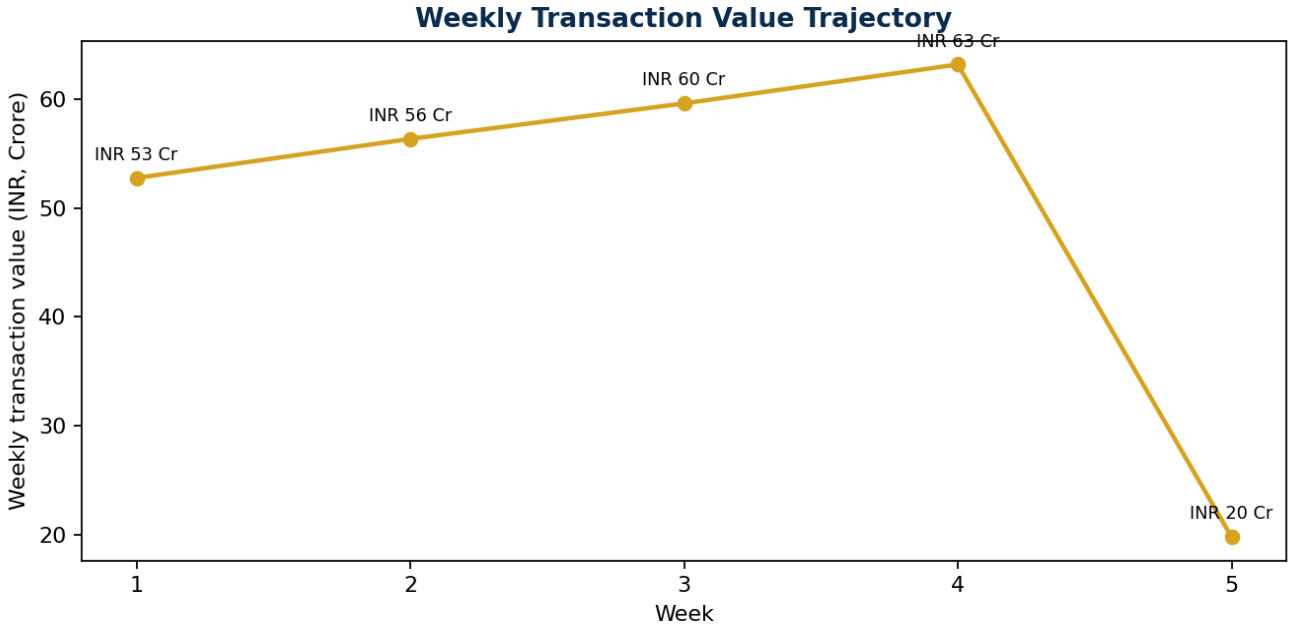


Figure 3. Weekly Gross Transaction Value Trajectory.

Week	Days	Order Count	Gross Value (INR Cr)	Value Share (%)
1	7	288,815	53	21.0%
2	7	307,840	56	22.4%
3	7	326,484	60	23.7%
4	7	352,003	63	25.1%
5	2	106,606	20	7.9%

Weeks 1 through 4 show a gradual build (from 21.0% to 25.1% of the period's value), consistent with the gentle 30-day growth trend layered into the daily simulation; Week 5 is a partial week (2 days) and should not be read as a demand contraction.

Deeper Insight — Partial-Week Truncation Artifact
Week 5's lower total is an analytical artifact of the 30-day collection window ending mid-week, not a genuine drop in transaction activity — the same caution that applies to any month- or week-end truncated reporting window.

3.0 Average Order Value & Binned Distribution Spread

Analyzing store-day average order values reveals the operational scale of purchasing activity — whether flows are dominated by small everyday purchases or large-ticket luxury/bulk orders.

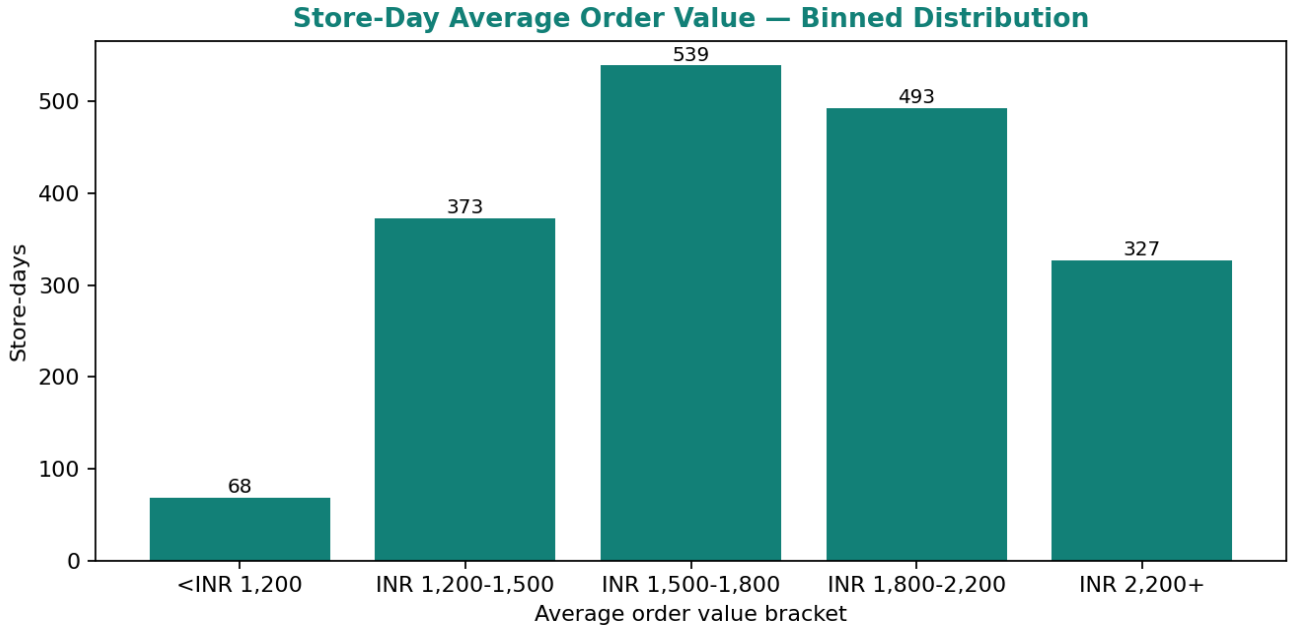


Figure 4. Store-Day Average Order Value — Binned Distribution.

Store-day average order values range from a minimum of INR 954 to a peak of INR 4,509. The overall **mean average-order-value is INR 1,823** and the **median is INR 1,759** — close to each other, indicating a moderately balanced distribution with an upper tail from luxury/bulk baskets.

Statistical Metric	Value (INR)	Commercial Interpretation
Minimum	954	Lowest store-day average order value recorded
First Quartile (Q1)	1,504	25% of store-days averaged this value or less
Median (50th percentile)	1,759	Midpoint of all store-day averages
Arithmetic Mean	1,823	Average order value per completed order
Third Quartile (Q3)	2,075	75% of store-days averaged this value or less
Maximum	4,509	Highest store-day average — luxury/bulk order activity
Interquartile Range (IQR)	571	Middle 50% spread of store-day average values

Commercial Business Recommendation
Stores whose average order value sits in the top bracket should be prioritized for dedicated optometrist appointments and premium eyewear styling delivery partnerships, since they are disproportionately handling high-ticket order activity.

4.0 Channel Tier Allocation & Infrastructure Capacity Dynamics

Analyzing which channel customers use to place orders reveals infrastructure load and guides capacity planning across the app, website, marketplace, and offline stores.

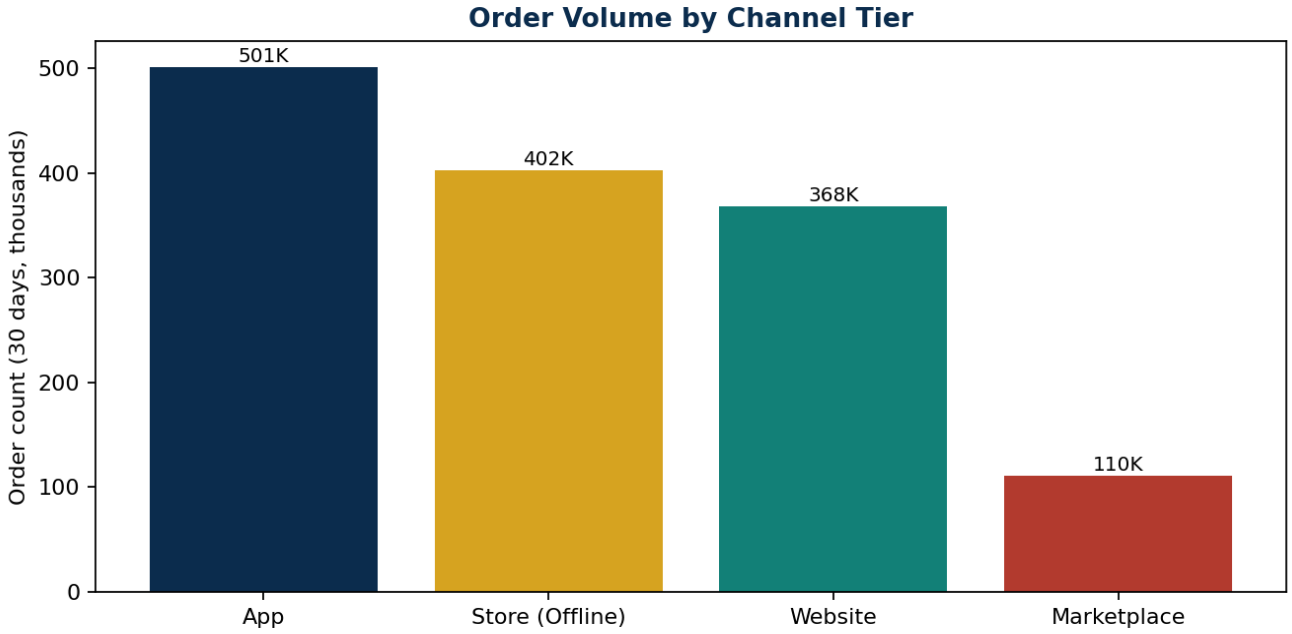


Figure 5. Order Volume by Channel Tier.

The **App** channel emerged as the highest-volume tier, capturing 500,923 orders (36.3% count share) and producing INR 78 crore in gross value.

Channel Tier	Order Count	Count Share (%)	Gross Value (INR Cr)	Avg Value/Order (INR)
App	500,923	36.3%	78	1,552
Store (Offline)	402,424	29.1%	94	2,324
Website	367,901	26.6%	65	1,758
Marketplace	110,500	8.0%	16	1,426

Deeper Insight — Digital vs. Physical Channel Split
App transactions carry the lowest average ticket size, consistent with small everyday purchases, while Offline Store and Website carry materially higher average values — reflecting in-store frame try-on and fitting behavior, and considered, higher-basket online purchases respectively.

5.0 Customer Segment Share & Relationship Revenue Contribution

Analyzing customer relationship segments clarifies which parts of the customer book drive value versus count, guiding loyalty-program and pricing strategy.

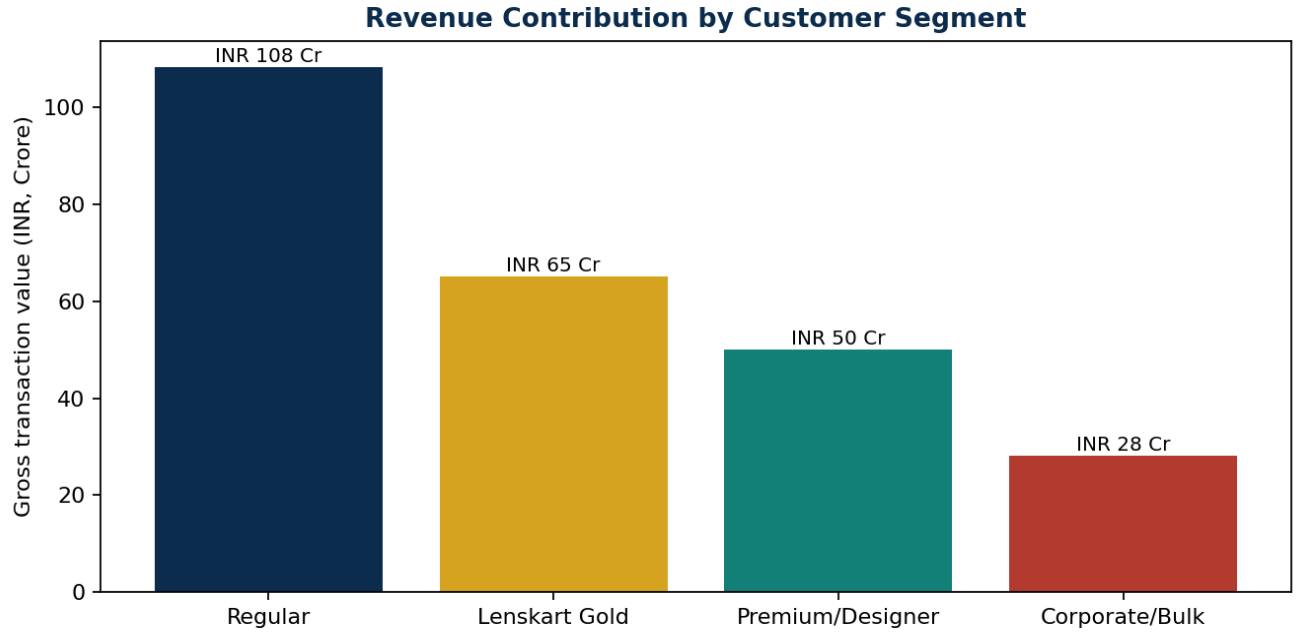


Figure 6. Revenue Contribution by Customer Segment.

Regular customers generated the largest single share of gross transaction value at INR 108 crore (43.1% share). Luxe and Corporate/Bulk segments, while smaller in order count, carry substantially higher average order values.

Segment	Order Count	Gross Value (INR Cr)	Value Share (%)	Avg Value/Order (INR)
Regular	757,680	108	43.1%	1,431
Lenskart Gold	376,934	65	25.9%	1,729
Premium/Designer	135,119	50	19.9%	3,699
Corporate/Bulk	112,015	28	11.2%	2,513

Relationship Management Recommendation
Luxe and Corporate/Bulk segments, despite a combined share well under half of order count, drive a disproportionate share of value — dedicated concierge servicing and priority store appointments for these segments protects a materially larger share of the value book than headcount alone would suggest.

6.0 Regional Demand Distribution & Geographic Concentration

Evaluating regional performance identifies primary geographic demand centers and highlights cross-region concentration.

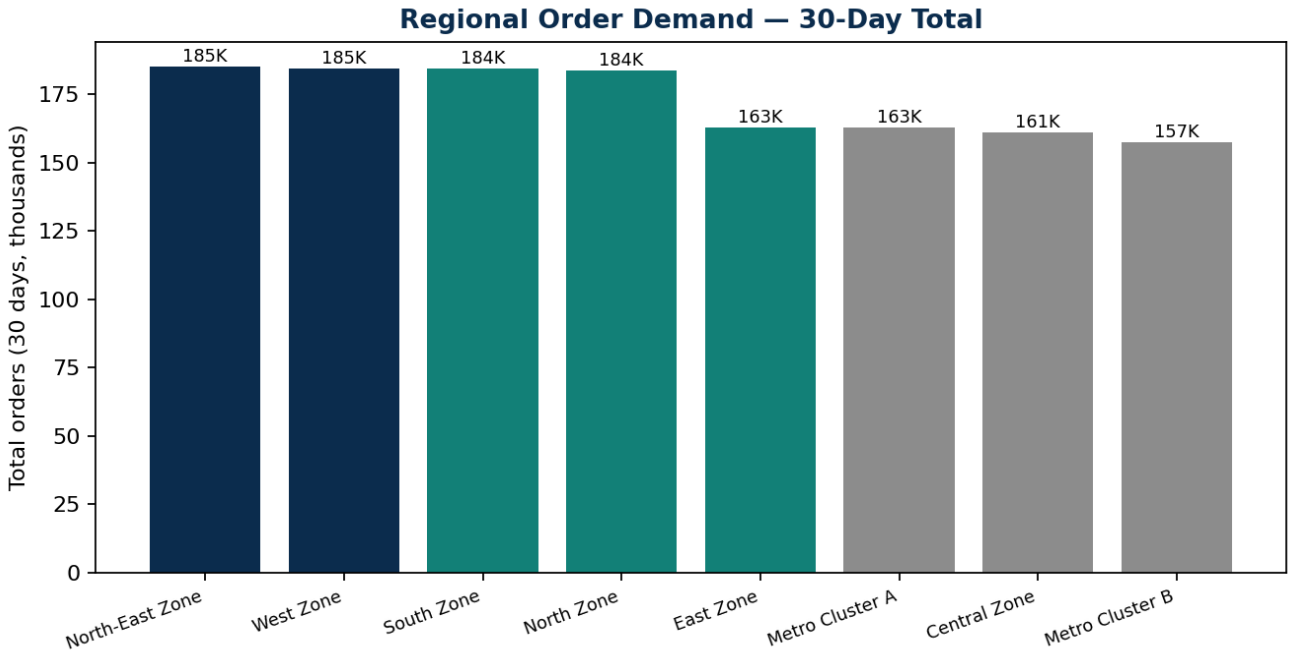


Figure 7. Regional Order Demand — 30-Day Total.

Transaction demand is led by **North-East Zone** at INR 33 crore (13.3% share), followed by **West Zone** at INR 34 crore (13.4% share). Together these two regions account for 26.7% of total transaction value.

Macro-Regional Clustering

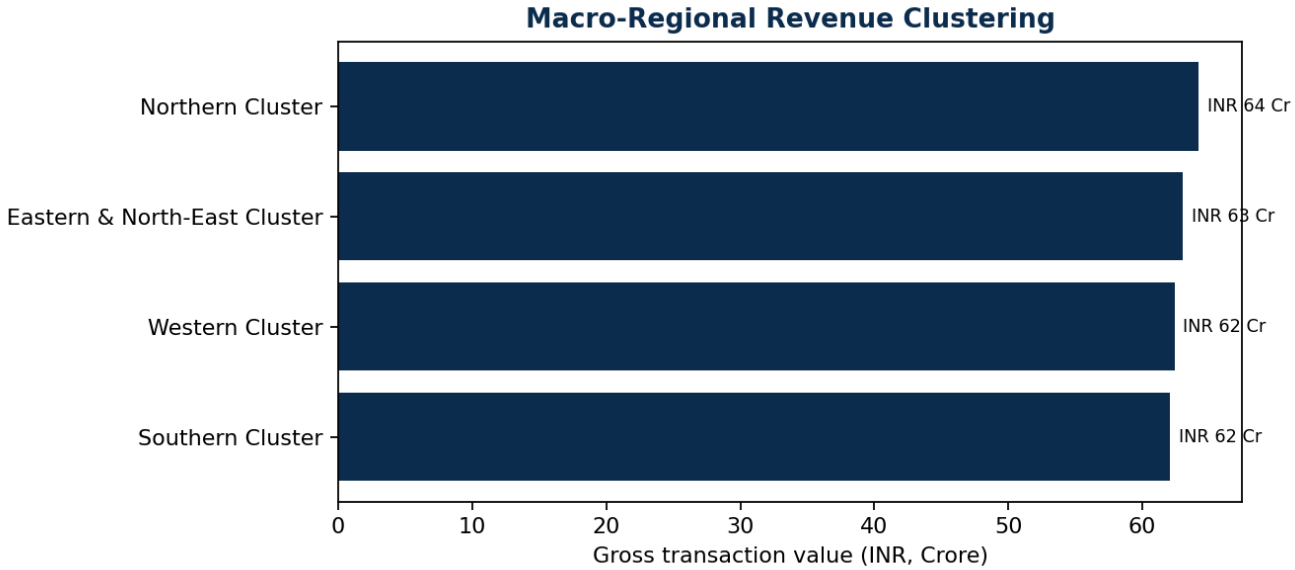


Figure 8. Macro-Regional Revenue Clustering.

Macro Cluster	Order Count	Gross Value (INR Cr)	Value Share (%)
Northern Cluster	346,533	64	25.5%
Eastern & North-East Cluster	347,871	63	25.0%
Western Cluster	345,620	62	24.8%
Southern Cluster	341,724	62	24.7%

Aggregating individual regions into macro-clusters provides strategic visibility into geographic revenue concentration for warehouse-network-investment and logistics planning.

7.0 Cross-Cutting Synthesis & Revenue Concentration Multipliers

Evaluating relationships across region and category together reveals the specific product-market clusters driving disproportionate revenue.

Rank	Region	Category	Order Count	Gross Value (INR Cr)
1	North Zone	Eyeglasses (Prescription)	31,283	9.0
2	South Zone	Eyeglasses (Prescription)	31,523	8.9
3	North-East Zone	Eyeglasses (Prescription)	31,568	8.7
4	West Zone	Eyeglasses (Prescription)	31,543	8.5
5	East Zone	Eyeglasses (Prescription)	27,718	7.9
6	Metro Cluster A	Eyeglasses (Prescription)	27,617	7.5
7	Metro Cluster B	Eyeglasses (Prescription)	26,647	7.4
8	Central Zone	Eyeglasses (Prescription)	27,502	7.3
9	West Zone	Sunglasses	27,679	6.3
10	South Zone	Sunglasses	27,617	6.2

Revenue Concentration: The High-Ticket Category Multiplier

The single strongest driver of top-line value is the concentration in high-ticket categories. Eyeglasses (Prescription) and Sunglasses together account for only 32.0% of order count but 43.7% of gross transaction value — a Pareto effect worth flagging for merchandising and inventory-investment prioritization.

Geographic and category dimensions represent the most stable analytical axes in the dataset, while weekly trends reflect expected operational seasonality and end-of-period collection boundaries.

8.0 Payment Rail / Settlement Preferences

Evaluating payment rails provides visibility into infrastructure dependency, gateway economics, and digital-payment adoption.

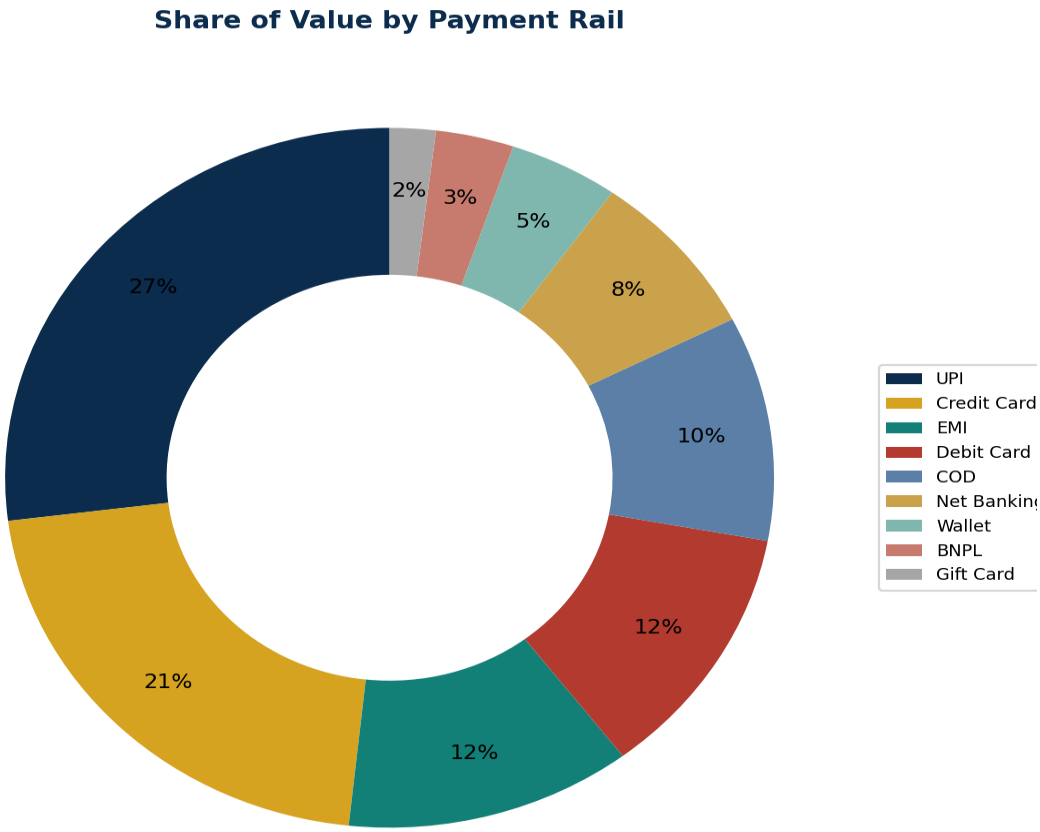


Figure 9. Share of Value by Payment Rail.

UPI dominates settlement value at 27.0% share, relative to its 32.1% share of order count.

Payment Rail	Order Count	Value Share (%)	Count Share (%)
UPI	444,136	27.0%	32.1%
Credit Card	217,223	21.3%	15.7%
EMI	74,595	12.1%	5.4%
Debit Card	171,485	11.7%	12.4%
COD	191,954	10.4%	13.9%
Net Banking	93,722	7.7%	6.8%
Wallet	111,561	4.6%	8.1%
BNPL	35,614	3.3%	2.6%

Gift Card	41,458	1.9%	3.0%

9.0 Order Success Rate & Reliability Analysis

Average success rate across all store-days: **97.4%**, clustered tightly. In a real system this view is most useful segmented, not aggregated — stores trending downward in rolling 7-day success rate are the earliest signal of a payment-gateway, warehouse-management, or courier-integration issue.

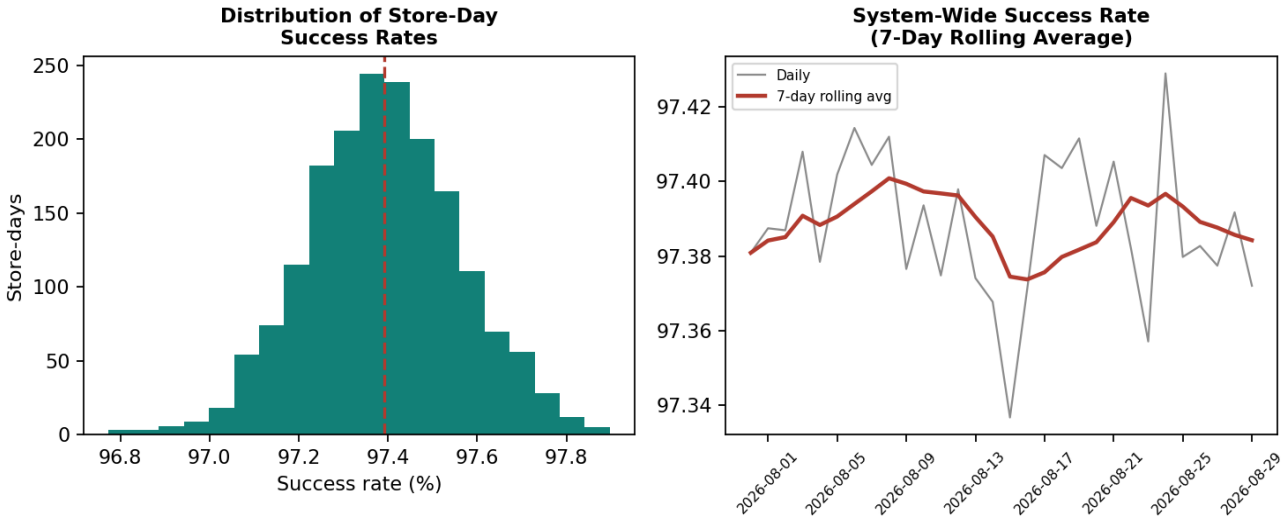


Figure 10. Store-Day Success Rate Distribution and 7-Day Rolling Average.

Deeper Insight — Reliability Drift
The 7-day rolling average shown here has held broadly steady over the observation window (97.38% → 97.38%) — worth a root-cause check before it compounds. Channels with high order volume but sub-97% success are the highest-risk combination, since volume amplifies the customer impact of every failed transaction.

10.0 Inventory / Stockout Risk (Stores)

Store frame-inventory reserves were bucketed into three tiers:

• **Adequate Reserve** — comfortable buffer for expected order demand

• **Low Reserve (Reorder Zone)** — below 35% of capacity, procurement reorder trigger

• **Critical — Stockout Risk** — below 15% of capacity, cross-referenced against top-store volume

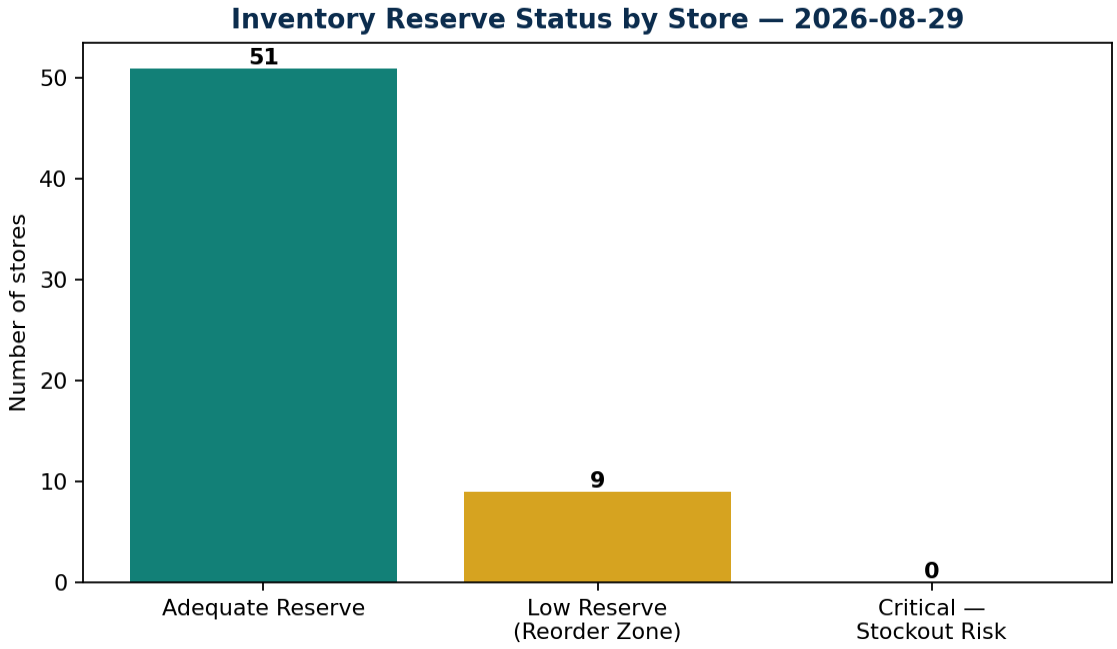


Figure 11. Inventory Reserve Status by Store — snapshot date.

Inventory-Logistics Recommendation
0 of 60 stores were flagged critical and 9 more sat in the reorder zone on the most recent day. Production dashboards should cross-tab inventory status against sell-through velocity to produce a “days of stock remaining” metric per SKU/store.

11.0 Volume, Value & Risk Elasticity Mechanics

Quantifying the mathematical relationships between order count, value, and reliability clarifies the key drivers of operational risk and top-line throughput.

Bivariate Variable Pair	Pearson Correlation (r)	Strength	Commercial Meaning
Order Count vs. Gross Value	+0.65	Moderate Positive	Stores with more daily orders substantially explained by order count, alongside category/seg
Order Count vs. Success Rate	-0.09	Negligible Negative	Order volume alone shows little linear relationship with success rate at this aggrega
Avg Value vs. Success Rate	+0.02	Negligible Positive	Average ticket size has little bearing on transaction reliability

Deeper Insight — Volume and Mix as Value Drivers

The positive correlation between order count and value (r = +0.65) shows that transaction count is an important lever for gross value, though category and segment mix add meaningful additional variation — consistent with a network where stores handle a broadly similar transaction mix but individual high-ticket categories (Designer/Luxury Eyewear, Computer & Blue-Light Glasses) still swing daily value.

12.0 Data Limitations, Scope & Assumptions

• **Dataset Scale:** The dataset contains ~1.38M simulated orders across 60 stores. While ideal for descriptive-analytics demonstration, it represents an illustrative sample rather than a real platform's total national volume.

• **Observation Period:** The dataset runs across 30 days with a partial final week, which should not be interpreted as an ongoing demand contraction.

• **Absence of Customer-Level Data:** The order records are store/region/category aggregates and do not include individual customer identifiers, KYC attributes, or return/refund records.

• **Descriptive Non-Causal Scope:** Findings such as the leading category's value leadership represent observed patterns within this dataset, not proof that category alone drives store-level risk.

• **Nominal Currency Reporting:** All financial figures reflect nominal INR as recorded, without adjustment for inflation or discounts/promotions.

Data Validation & Governance

All aggregated rows were verified for schema consistency, non-negative values, and mathematical consistency (count x avg value = total value). Zero anomalies or negative values were found.

13.0 Next Steps: Move From Descriptive → Predictive Analytics

Descriptive analytics explains past order outcomes. The next analytical step is leveraging these empirical patterns to train predictive models that forecast future demand and risk.

Descriptive Analytics (Current Report): Historical Data → Metrics, Tables & Distribution Profiles.
Predictive Analytics (Next Phase): Historical Baselines → Machine Learning Models & Demand/Risk Forecasts.
Prescriptive Analytics (Advanced Phase): Predictions → Automated Inventory Replenishment, Dynamic Fraud Rules & Delivery Routing.

Predictive Target & Objective	Candidate ML Models	Key Input Features
Demand forecasting per SKU/store	ARIMA, SARIMAX, Prophet, LSTM	Historical order volume, calendar effects, sale-event dates
Real-time fraud / anomaly detection	Isolation Forests, Autoencoders, XGBoost	Order velocity, amount deviation, channel, time-of-day
Success-rate degradation early warning	Change-point detection, Random Forest	Rolling success rate, store load, gateway latency
Customer cohort & retention modeling	Gradient Boosted Regressors, survival models	Order recency, frequency, segment, channel mix

Happy to build out any of these next, or swap in a real dataset (order-management export, a data warehouse connection, etc.) if you have one — the same pipeline and chart structure above would apply directly.

14.0 Summary of Key Metrics & Executive Scorecard

Metric Area	Top Segment / Value	Metric Value	Commercial Takeaway
Gross Transaction Value	Entire Dataset	INR 252 Cr	Total value across 30 days
Total Order Volume	All Categories	1.38M orders	Aggregate volume across 10 categories
Top Category by Value	Eyeglasses (Prescription)	25.9% share	Leading value contributor
Top Category by Volume	Eyeglasses (Prescription)	17.0% share	Leading count contributor
Top Geographic Region	North-East Zone	13.3% share	Largest single-region market
Top Customer Segment	Regular	43.1% share	Largest value-contributing segment
Dominant Channel	App	36.3% share	Primary order-initiation channel
Average Success Rate	All Stores	97.4%	System-wide transaction reliability
Stores at Inventory Risk	Low + Critical Reserve	9 of 60	Inventory-logistics attention needed

References & Methodology Sources

1. RBI (Reserve Bank of India). Payment System Indicators & Digital Payment Trends. Official RBI publications.
2. NPCI (National Payments Corporation of India). UPI, IMPS & RuPay Transaction Statistics.
3. McKinsey & Company. Global Eyewear & Optical Retail Analytics: Descriptive Foundations to Real-Time Predictive AI. McKinsey Insights.
4. Provost, F., & Fawcett, T. Data Science for Business: Fundamental Principles of Data Mining & Analytic Thinking. O'Reilly Media.
5. Gartner Research. E-commerce Transaction Volumes & Digital Channel Adoption Trends.

Dataset Citation: Lenskart Daily Transaction Dataset (60 stores x 10 categories x 8 regions x 30 days, ~1.38M simulated orders): synthetic project dataset recording Date, Store_ID, Region, Category, Order_Count, Total_Value, Success_Rate, Declined_Count, and Inventory_Reserve_Pct.

Technical Appendix — Schema & Formulas

Appendix A: Database Schema & Field Data Dictionary

Field Name	Data Type	Sample Value	Description
date	Date	2026-07-31	Order date (YYYY-MM-DD)
store_id	Categorical	ST033	Store identifier (60 unique stores)
region	Categorical	South Zone	Geographic region (8 unique zones)
category	Categorical	Skincare	Product category (10 distinct categories)
order_count	Integer	1,204	Number of orders in this row
total_value_inr	Float	1,024,500.00	Aggregate value in INR
success_rate_pct	Float	97.5	Share of orders completed successfully
declined_count	Integer	12	Number of declined/failed orders
inventory_reserve_pct	Float	0.55	Store inventory reserve as a fraction of capacity

Appendix B: Aggregation Formulations

- Gross Transaction Value: Sum(total_value_inr) across all rows
- Average Value per Order: Gross Value / Total Order Count
- Category / Segment / Region Value Share (%): (Category Value / Total Gross Value) x 100
- Success Rate (weighted): Weighted average of success_rate_pct, weighted by order_count

Glossary of E-commerce Analytics Terms

Term	Definition & Commercial Context
Descriptive Analytics	Summarizing and visualizing historical order data to understand past operational performance without predictive modeling.
Average Order Value (AOV)	An e-commerce operations KPI measuring the average value moved each time an order completes.
Revenue/Value Concentration	A distribution pattern where a small number of categories or regions generate a disproportionate share of total value.
Interquartile Range (IQR)	The statistical spread between the 25th (Q1) and 75th (Q3) percentiles, capturing the middle 50% of values.
Success Rate	The share of initiated orders that complete successfully, net of declines and technical failures.
Inventory / Stockout Risk	The risk that a store runs below the frame-inventory reserve needed to meet expected order demand.
Pearson Correlation (r)	A statistical coefficient measuring the linear relationship between two numerical variables.
Cross-Dimensional Decomposition	Simultaneously grouping order data across two categorical dimensions (e.g. Region x Category) to evaluate granular performance
Predictive Transition	Using historical order patterns to engineer features and train supervised ML models for demand or risk forecasting.

Report Conclusion

This concludes the Lenskart Daily Transaction Descriptive Analytics Report. All analyses, tables, charts, and technical appendices are fully documented and validated for review.